

Sagar Fab International Company

Full Stack Developer Assignment - Product Thinking Report

Product Design & Technical Decisions

Role: Full Stack Developer (Ruby on Rails + Flutter)

Date: July 2, 2026

1. What happens when the library is empty?

A blank black void feels broken, confusing, or incomplete. When the library contains no books, and the user is in Bookshelf View, the application continues to render the physical mahogany background and empty wooden shelves. On top of this, we overlay a soft, translucent dark card prompting the user to upload their first file and providing a direct quick-action button, ensuring the design language stays intact while offering a clear call-to-action.

2. What happens when upload fails?

Losing typed details during an upload error leads to a frustrating user experience. During an upload, all input parameters are locked, and a loader displays. If the API request fails, the dialog remains open with the user's typed titles and authors preserved, presenting a red warning description. This allows the user to immediately select another file or re-submit without having to start over.

3. What happens when the ebook file is too large?

Uploading oversized documents wastes battery and network bandwidth only to fail at the server. We enforce a two-tier validation approach. First, the client-side FilePicker immediately checks the document size. If it exceeds 50MB, it blocks the upload action and presents an inline warning. Second, the backend Rails model performs database validations, refusing attachments above 50MB as a fallback.

4. What happens when search has no results?

If a query returns no matching items, we render a clean empty result indicator displaying: "No Results Found. No ebooks match your query [Keyword]". The horizontal "Continue Reading" slider is also hidden during searches to declutter the viewport and let the user focus entirely on search results.

5. What should the user see while data is loading?

We use context-specific loading shimmers and indicators. The home screen shows a centered spinner while loading books. The upload dialog uses a fullscreen progress overlay with status texts: "Uploading Ebook... Parsing PDF metadata & saving file". Active file downloads display a real-time slider progress bar indicating the exact download percentages.

6. How should errors be displayed?

Errors are separated by severity to avoid blocking navigation. Severe startup issues (like a disconnected Rails

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instance) replace the library list with a full-page error state and a "Try Again" refresh button. Transient action errors (like a failed download or delete request) present red Snackbar alerts at the bottom of the screen, notifying the user without tearing down their current screen.

7. How should delete confirmation work?

Deleting a book is permanent, wiping database rows and active storage binaries. We display a modal AlertDialog highlighting the target book title: "Are you sure you want to permanently delete [Title] from the library? This cannot be undone." The confirmation trigger is styled in warning red, and Cancel is set as the default option.

8. How does the app remain simple and easy to use?

By removing cognitive load. Debounced search handles updates seamlessly without forcing manual submission taps. Automatic PDF metadata parsing extracts title and author information behind the scenes, making manual entry optional. Local page memory automatically bookmarks reading progress, and local download checks allow offline reading without any manual configurations.